

Likelihood-Ranked Selection in Self-Consuming Diffusion: A W_2 Negative Result and a Narrow Minority-Retention Regime

Muhammad Alamzeb
Independent researcher
shayankhanmahar@gmail.com

Abstract

Self-consuming training loops can induce *model collapse*. We study unlabeled likelihood ranking of synthetics in self-consuming DDPM training on imbalanced mixtures (workshop-scale; not ImageNet diffusion). **Primary:** top- k does *not* improve sliced W_2 vs. random- k at generation $g=1$ ($\Delta W_2 > 0$ on 38/38 cells; bootstrap 95% CI excludes zero). **Secondary:** after falsifying an earlier “false stability” hypothesis, minority retention r shows bottom>rand>top mainly at moderate imbalance $\rho=5$ (exploratory $n=10$, 10/10, exact $p=0.002$; held-out seeds 10–14: 4/5, CIs still exclude zero for both contrasts). Full order is weaker at $\rho=7$ (6/10) and $\rho=10$ (4/13). Code: <https://github.com/muhammadalahamzeb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse* [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024] and filtering synthetics [Feng et al., 2024, Cai et al., 2025]. When the only score is the generator’s own likelihood, how should one select synthetics? We isolate an interventional selection policy at matched synthetic count and matched real-mix α .

Caveats that apply throughout: this is a workshop empirical note on controllable mixtures; the retention ordering is post-hoc relative to an earlier falsified hypothesis (Section 6); we do not claim ImageNet-scale generality.

Contributions. (1) Primary powered W_2 negative result for top- k at $g=1$ ($n=38$ cells). (2) Stratified retention at $\rho \in \{5, 7, 10\}$, with a labeled held-out seed block at $\rho=5$. (3) A lightweight account of why the full order weakens as ρ grows (minority-mass floor). (4) Explicit contrast to siloed biased-verifier selection [Qiao et al., 2026].

2 Related work

Shumailov et al. [2023], Alemohammad et al. [2023] document collapse and quality-biased sampling (*MAD*). Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Feng et al. [2024] study task verification of labeled synth; Cai et al. [2025] (*LSF*) filter via latent probes. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we intervene on training-set selection.

Table 1: Differentiation vs. Qiao et al. [2026] (sample-selection bias precipitating collapse).

	Qiao et al. [2026]	This note
Verifier	Biased <i>local real</i> reference (silos)	Generator’s <i>own</i> likelihood proxy
Setting	Multi-party / privacy silos	Single self-consuming loop
Intervention	Local vs. collaborative Wasserstein proxy	Matched top / rand / bottom- k
Primary diversity metric	Variance / diversity decay	Mode masses r + sliced W_2
Main message	Local filtering can accelerate collapse	Top- k harms W_2 ; r -order is ρ -dependent

3 Method

3.1 Notation

Table 2: Symbols used throughout.

Symbol	Meaning
ρ	Train imbalance: majority mode weight boost before renormalization
g	Self-consuming generation index ($g=0$ is the first model on real data)
r	Minority-mass retention $m^{(1)}/m^{(0)}$ at consecutive generations
α	Fraction of each train set drawn from a fixed real buffer
k_{frac}	Selection-pool size as a fraction of N synthetics
$s(x)$	Denoising-MSE likelihood proxy (lower = higher likelihood)

3.2 Self-consuming loop

Train DDPM $f^{(g)}$ on $\mathcal{D}^{(g)}$, sample synthetics, mix fixed real buffer (fraction α) with selected synthetics.

3.3 Likelihood proxy and policies

Score $s(x)$: mean denoising MSE over a t -grid (lower = higher likelihood).

- **top- k / bottom- k / random- k :** pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}}N \rfloor)$ by lowest / highest / uniform scores; then draw n_{syn} and mix with α real.
- **mix:** same α , but synthetic portion is a uniform subsample of all N synthetics (no rank filter).
- **replace:** pure synthetic.

Thus mix $\neq$ random- k .

3.4 Metrics

Nearest-mean mode masses; minority-mean m ; retention $r = m^{(1)}/m^{(0)}$ (allowed > 1). Sliced W_2 : mean squared 1D Wasserstein over **24** projections; ≤ 400 synth and ≤ 400 held-out real points. Hypothesis tests: bootstrap 95% CIs of the mean paired Δ ; Wilcoxon signed-rank with exact sign enumeration for $n \leq 16$ and a normal approximation for larger n . Cells with sliced $W_2 > 10$ (pathological late-generation collapse) are excluded from W_2 tables only.

Table 3: $\Delta W_2(\text{top} - \text{rand})$ by generation (positive = top worse). Bootstrap 95% CI; Wilcoxon; Holm across generations. One $g=5$ cell excluded ($W_2 > 10$).

g	Mean	Std	d_z	95% CI	$\#\{> 0\}$	p (Holm)
1	+0.087	0.075	1.16	[0.064, 0.112]	38/38	$< 10^{-6}$
2	+0.043	0.044	0.96	[0.029, 0.057]	32/38	5×10^{-6} (1.5×10^{-5})
3	+0.038	0.039	0.98	[0.026, 0.051]	36/38	1×10^{-6} (3×10^{-6})
4	+0.022	0.044	0.49	[0.008, 0.035]	29/38	0.00029
5	+0.027	0.030	0.91	[0.018, 0.037]	31/37	6×10^{-6} (1.5×10^{-5})

Table 4: Per- ρ mean r and full-order counts (generation $g=1$).

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	15	0.604	0.350	0.128	14/15	15/15 / 14/15
7	10	0.372	0.248	0.113	6/10	9/10 / 8/10
10	13	0.248	0.183	0.120	4/13	10/13 / 8/13

4 Experiments

4.1 Design ($n=38$ cells)

$K=4$, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Log: `w2_fs.jsonl`. Seeds: $\rho=5$ uses 0–14 (0–9 exploratory; 10–14 held-out); $\rho=7$ uses 0–9; $\rho=10$ uses 0–12.

4.2 Sliced W_2 (primary)

4.3 Retention (secondary; stratified)

Held-out seeds 10–14 were not used during exploratory search; they replicate the *direction* (CIs exclude zero; full order 4/5) but, at $n=5$, exact Wilcoxon remains floored for bottom–rand. At higher ρ , top still tends to underperform random, but the full three-way order is no longer reliable.

Illustrative α / k_{frac} ablations ($n=2$) and 8D GMM / Digits-PCA transfers ($n=3$) are in Appendix A. Digits pixel-DDPM and Digits-CVAE are reported there as preliminary/negative, not as main transfer claims.

5 Analysis: why the order weakens with ρ

Top- k keeps low- $s(x)$ synthetics, which under an imbalanced teacher are disproportionately majority-mode. As ρ grows, minority mass at $g=0$ already shrinks, so the ranked pool contains few minority candidates to prefer or discard: top/random/bottom selection become more similar, r contrasts shrink, and the full bottom>rand>top pattern becomes seed-noisy—a minority-mass floor, not a contradiction of the $\rho=5$ regime. Aggregate W_2 remains a poor minority monitor: top- k is worse on W_2 at $g=1$ across the whole matrix even where the three-way r -order fails.

Table 5: Paired retention contrasts vs. random- k (bootstrap 95% CI; Wilcoxon).

Stratum	Contrast	Mean Δ	d_z	95% CI	p
$\rho=5$ (all)	top-rand	-0.222	-1.76	$[-0.28, -0.16]$	0.0001
$\rho=5$ (all)	bottom-rand	+0.254	+1.96	$[0.19, 0.32]$	0.0001
$\rho=5$ expl. 0-9	top-rand	-0.225	-1.82	$[-0.29, -0.15]$	0.002
$\rho=5$ expl. 0-9	bottom-rand	+0.285	+2.08	$[0.21, 0.37]$	0.002
$\rho=5$ held-out 10-14	top-rand	-0.216	-1.49	$[-0.32, -0.09]$	0.125
$\rho=5$ held-out 10-14	bottom-rand	+0.191	+2.01	$[0.11, 0.25]$	0.0625
$\rho=7$	top-rand	-0.135	-1.58	$[-0.19, -0.08]$	0.0039
$\rho=7$	bottom-rand	+0.124	+0.88	$[0.05, 0.21]$	0.0195
$\rho=10$	top-rand	-0.063	-0.71	$[-0.11, -0.02]$	0.037
$\rho=10$	bottom-rand	+0.065	+0.44	$[-0.00, 0.15]$	0.204

6 Limitations

Scale. 2D/8D GMMs only for primary claims; no CIFAR/ImageNet UNet or DiT result. **Process.** Documented exploratory search (>100 killed directions in `docs/KILL_LOG.md`) precedes this topic; within-topic, the original false-stability hypothesis was falsified and the retention claim revised (HARKing risk). Mitigation: W_2 primary; stratified ρ ; held-out seeds 10-14 at $\rho=5$; bootstrap CIs. **Other.** α/k_{frac} ablations are $n=2$; Holm within W_2 generations and within paired $\rho=5$ contrasts.

7 Conclusion

Top- k likelihood selection does not improve sliced W_2 at $g=1$ in this matrix. Minority-retention ordering is strongest at moderate imbalance and weakens as minority mass floors—a mechanism-level boundary, not a universal law.

References

- Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. Self-consuming generative models go MAD, 2023.
- Zhongteng Cai, Yaxuan Wang, Yang Liu, and Xueru Zhang. Stabilizing self-consuming diffusion models with latent space filtering, 2025.
- Yunzhen Feng, Elvis Dohmatob, Pu Yang, Francois Charton, and Julia Kempe. Beyond model collapse: Scaling up with synthesized data requires verification, 2024.
- Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Dhruv Pai, Henry Sleight, John Hughes, Tomasz Korbak, Rajashree Agrawal, Andrey Gromov, Daniel A. Roberts, Diyi Yang, David Donoho, and Sanmi Koyejo. Is model collapse inevitable? breaking the curse of recursion by accumulating real and synthetic data, 2024.
- Peizhuo Li, Emre Aksan, Alexandru-Eugen Ichim, Thabo Beeler, and Olga Sorkine-Hornung. Diversify diffusion with temperature sampling and variance-corrective time shifting, 2026.

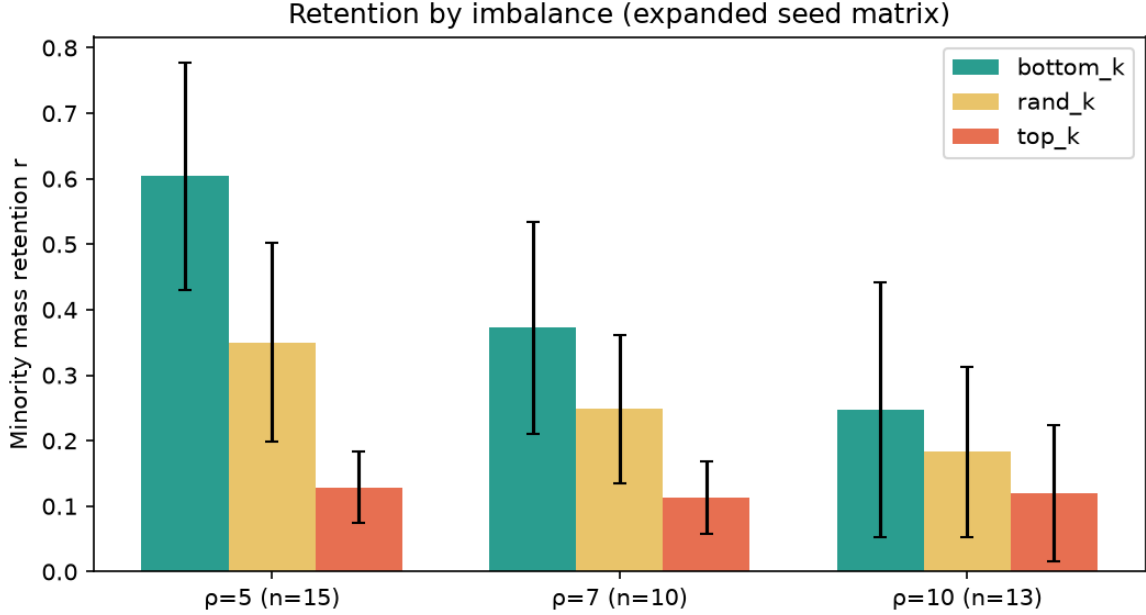


Figure 1: Retention r by ρ (means $\pm$ sd over seeds).

Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, and Yan Pang. When sample selection bias precipitates model collapse, 2026. ICML 2026.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget, 2023.

Yanbo Xu, Yu Wu, Sungjae Park, Zhizhou Zhou, and Shubham Tulsiani. Temporal score rescaling for temperature sampling in diffusion and flow models, 2025.

A Extra ablations and preliminary image checks

α and k_{frac} ($n=2$, illustrative). At $\rho=5$, order holds 1/2 at $\alpha \in \{0.25, 0.50\}$ and 0/2 at $\alpha=0.75$; $k_{\text{frac}}=0.25$ coincides with 0.5 when n_{syn} binds, and 0.75 breaks full order (0/2). Default rows of the two ablations are the same experimental cell.

Directional transfers ($n=3$). 8D GMM and Digits-PCA: full order 3/3 (exact p floor 0.25).

Preliminary / negative: Digits pixel-space DDPM full order 1/3; Digits-CVAE inconclusive. These do not support a main-text transfer claim.

B Reproducibility

python scripts/reproduce_w2.py; python experiments/scripts/make_submission_stats.py.
Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.